Lab 5: Analog Signal Conditioning and Filter Design for Mechatronic Systems

Objectives: Passive and Active Analog Filters Design and Implementation.

Timeline

This lab spans two weeks. Complete the pre-lab tasks **before the first lab session**. Review all reading materials **during the two weeks but before attending each lab session**. Do not wait until the lab sessions to complete these preparations.

Pre-Lab Tasks

Due: Before the first lab session. Note: You are not required to upload anything to D2L. This pre-lab is for your own practice and preparation only.

Complete the following tasks:

- Read about Low Pass Filters `https://analogcircuitdesign.com/low-pass-filters/`
- Read about High pass filters `https://analogcircuitdesign.com/high-pass-filters/`
- Read about Analog Filters `https://analogcircuitdesign.com/introduction-to-analog-filters/`
- Familiarize yourself with tools to draw circuits: `https://www.circuit-diagram.org`

Required Reading

Complete during the two-week lab period, before attending lab sessions. Posted to D2L

- Analog Filters – 19 pages – From Chapter 4, Book: *Mechatronics An Integrated Approach by Clarence W. De Silva.* This covers passive and active analog filters, including lowpass, highpass, band pass, and bandreject filters.

Mechatronics with Experiments **by Cetinkunt, Sabri. (Online Access Through UCalgary Library):**

- Section 5.6 – Pages 301 to 308–Topics includes Low Pass Filter Op-Amp, High Pass Filter Op-Amp, Band Pass Filter Op-Amp, Band Reject Filter Op-Amp, Instrumentation Op-Amp excluding Current-to-Voltage Converter and Voltage-to-Current Converter. This covers active filters. [1]
- Read about Difference between Bessel, Butterworth, Chebyshev Filters: `https://www.ti.com/lit/an/slyt681/slyt681.pdf?ts=1762328415714` and `https://analogcircuitdesign.com/butterworth-and-chebyshev-filters/`

[1] Please note that since this book is available through the University of Calgary's online library, screenshots will not be uploaded to D2L in order to comply with copyright regulations.

- Read about Bandstop filters and Notch filters
  `https://analogcircuitdesign.com/band-stop-filter-and-notch-filter/`
- Read about Bandpass Filters
  `https://analogcircuitdesign.com/bandpass-filters/`
- After completing the readings above, proceed to explore different filter types, specifications, and component selections via the following online tool
  `https://tools.analog.com/en/filterwizard`